

BOXER User Guide
Version 5.0
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Historical background

The original version of the Boxer program was written by Paolo Gasperini, in the month of April 1999, with the contribution of Filippo Bernardini, Gianluca Valensise (who also suggested the name) and Graziano Ferrari. It consisted of a reorganization and standardization of various Fortran codes developed since 1993, which had been used to compute synthetic earthquake parameters of the *Catalogo dei Forti Terremoti in Italia dal 461 a.C. al 1990* (CFTI Versions 1 and 2) (Boschi *et al.* 1995; 1997), the *Catalogo Parametrico dei terremoti Italiani (CPTI)* (CPTI Working group, 1999) and to make the computations of Gasperini *et al.* (1999). The successive public releases of the program up to version 3.3 also took advantage of important improvements made by Gianfranco Vannucci (particularly regarding the production of MapInfo and GMT output files to plot results). Versions 3.2 and 3.3 of the code have been used to compute the parameters of the CFTI versions 3 and 4 (Boschi *et al.*, 2000; Guidoboni *et al.*, 2007) and of CPTI version 04 (CPTI Working group, 2004). The latest released version 3.3 also includes a procedure to calibrate magnitude coefficients, based on a set of macroseismic data with known magnitude, which have been employed in a benchmark among different codes, within the ambit of the NERIES Network Project of the E.U. Framework Program 6. A description of the procedure to compute the azimuth and to draw the “boxes” representing the seismogenic sources can be found in Gasperini *et al.* (1999). More details on the computation of epicentre and magnitude are given by Gasperini and Ferrari (1995; 1997; 2000).

An updated version (4.0) of the code was developed by Gasperini *et al.* (2010) with the contribution of Gianfranco Vannucci, Daniele Tripone ed Enzo Boschi. In particular, such release introduced new methods to compute the location and the magnitude as well as to estimate parameter uncertainties by the bootstrap approach (Efron and Tibshirani, 1986). All these new improvements of version 4.0 of the code was developed by Gianfranco Vannucci and Paolo Gasperini. Daniele Tripone also developed a window-based user-friendly interface to setup the input parameters and to post-process the results that was included in the package.

The present updated version (5.0) of the code was developed by Gianfranco Vannucci and Paolo Gasperini Vannucci and Gasperini, (2025) and consists of a complete revision of the Fortran source code so that to complain with the GNU Fortran compiler. It can be compiled and run, in operating systems other than Windows and includes new features, responding to user needs gathered over the past 15 years, like the use of symbol-delimited input files, or the introduction of new computational approaches. We warmly invite the users to report to the authors (by emails to gianfranco.vannucci@ingv.it and paolo.gasperini@unibo.it) any comments and suggestions or requests for new features that might be included in future updates.

The main new feature is the increase in the number of platforms on which the code can be used, no longer limited to the MS-Windows environment but also extended to MacOS/Darwin and Linux/Ubuntu environments. To achieve this, the original code has been adapted to be compiled by the GNU Fortran (gfortran) compiler, both in Intel and ARM processors. The executable codes have different extensions for MS-Windows, MacOS/Darwin, and Linux/Ubuntu environments (.exe, .e, .u, respectively). About the correspondence of the results in the various system environments see Vannucci and Gasperini (2026).

Distribution kit and installation

The executable software for MS-Windows, MacOS/Darwin and Linux/Ubuntu environments can be downloaded from: <https://emidius.mi.ingv.it/boxer/> and <https://github.com/gfvan23/BOXER>. The

auto-extracting executable kit files includes the following files:

- BOXER5dg_intel_WIN.exe (program executable for Intel x86_64 processors, for Windows)
- BOXER5dg_intel_MAC.e (program executable for Intel x86_64 processors, for MacOS/Darwin)
- BOXER5dg_intel_LNX.u (program executable for Intel x86_64 processors, for Linux/Ubuntu)
- BOXER5dg_arm_WIN.exe (program executable for ARM64 processors, for Windows)
- BOXER5dg_arm_MAC.e (program executable for ARM64 processors, for MacOS/Darwin)
- BOXER5dg_arm_LNX.u (program executable for ARM64 processors, for Linux/Ubuntu)
- inpparm.dat (Example option card file)
- input.txt (Example intensity data file)
- in folder OUTPUT_intel_WIN
 - outfull.dat (Example full output file)
 - outsummary.dat (Example summary output file, old format)
 - outsummary_new.dat (Example summary output file, formatted text)
 - outsummary_new.xls (Example summary output file, tab separated text)
- polyline.txt (Example of Polyline)
- boxer5_package.f (Fortran source of key routine computing location, magnitude and fault orientation)
- BOXER5_manual.pdf (this file)

To install BOXER 5 code, copy and run the BOXER5_kit.zip file in a disk folder.

Input and output

The BOXER code reads user directives (“cards”) for data processing from the file inpparm.dat. These cards indicate tasks to perform, options or specific parameters to be used in computations. The inpparm.dat file must reside in the same folder of the code executable file. Earthquake intensity data are read from another file, which name and format can be specified by the user (see below). For each earthquake, an identification record (including date, epicentral area, instrumental magnitude, etc.) as well as the intensity observations can be read from this file. The folder and/or the device where the input and output files reside can be specified by cards. The code execution produces four output files: outfull.dat, including all the details of computations in a easily readable ascii format; outsummary.dat, kept for compatibility with previous versions, containing a record for each earthquake and including only source parameters computed by methods available up to version 3.3; outsummary_new.dat and outsummary_new.xls, containing a record for each location method (up to 7 records for each earthquake) and including all of the parameters computed by methods as formatted ascii text and tab-delimited text files respectively (see description of the fields below). The procedure for the calibration of new coefficients for computing magnitude by the isoseismal areas method (COMPCOE and COMPCATT cards) produces the file outparm.dat, which is a copy of file inpparm.dat with the new computed coefficients added at end. Other files are also generated when the option cards MAPINFO and GMT are present. They are stored within a subfolder tree described below.

Changes with respect to previous versions

In release 4.0, some minor changes, mainly concerning the computation of uncertainties, affected the location and magnitude methods used in previous versions (see *Gasperini et al.* (2010)). In particular, the uncertainties of epicentral coordinates computed by the barycentre location algorithm and of the fault orientation were corrected so that to represent correctly the variability of averages rather than that of the observations. Furthermore, some computational details of the Rayleigh and Kuiper tests, to assess the uniformity of the azimuthal distribution of intensity data used to compute the fault orientation, have been changed according to the recent literature. The latter changes made the Kuiper test more restrictive and might lead to discard orientations that were previously accepted.

However, as the Kuiper test does not take into account correctly the weight given to each intensity observation, *Gasperini et al.* (2010) suggested to consider its results only if the Rayleigh test does not reject the uniformity hypothesis. In the practice, this means discarding the orientation only if both tests do not reject the uniformity hypothesis ($s.l. > 0.1$).

Since release 4.0, alternative location and magnitude methods can be selected by including the ATTLOC and ATTMAG option cards in the input deck. One or more of the 6 the location algorithms described by *Gasperini et al.* (2010) can be selected, but the barycentre method is also computed in all cases. The BTSTRP option card activates the computation of bootstrap uncertainties, while the MUSSONDEP card activates the computation of depth according to the method proposed by Musson (1996) as described in *Gasperini et al.* (2010). Further option cards INPFOLDER, OUTFOLDER, PATHCD, ELFACT, ATTLCOE, and M-IECATT allow respectively to set the folders containing input and output files (excluding inpparm.dat), to change the root device and/or folder of input and output files (excluding inpparm.dat), to set the confidence level of uncertainty ellipses or ellipsoids, to set the coefficients of the log-linear attenuation equation (*Pasolini et al.* 2008) used by new location and magnitude methods, and to set the coefficient of the linear relationship between the intensity at the epicentre I_E and moment magnitude M_w . The new output files outsummary_new.dat and outsummary_new.xls report all the parameters computed for each earthquake and/or location method.

The new code also implements a procedure to recover the results of a program run that have been interrupted for any reason. To recover the job, starting from the earthquake that was being computed at the moment of the interruption, the user must answer 'y' to the program prompt. Any other answer makes the program to restart from the beginning of the input file.

Limits: the maximum number of earthquakes to process as a catalogue is 21,000, with a maximum number of MDPs per earthquake of 2,500.

Release 5

The new features in release 5 are clearly indicated by the new cards that have been added. These cards also address user requests and feedback aimed at improving and simplifying the use of the code.

The most significant change was the rewriting of the code in gfortran, including all the calculation routines used in the code to produce executables for Windows (WIN), MacOS/Darwin (MAC), and Linux/Ubuntu (LNX) environments.

Option and parameter cards in file inpparm.dat

The order of occurrence of the cards in the input deck is usually irrelevant excepting for the COMPCOE and COMPCOA cards (which must be the at the beginning of deck) and the multiple appearance of the same option card for which only the parameters of the last one are retained.

Each card has a "name" field (from columns 1 to 10) and a "parameter" field (from columns 11 to 200). The MAGCOEFF card is followed by one or more additional parameter cards.

The name of the card must be written starting from column 1, without abbreviations, and using either upper- or lower-case characters. The percent and number signs ("% o "#") in column 1 indicates a comment card that is skipped by the program (but printed in the outfull.dat file). Only upper- and lower-case letters, numbers and "-" are valid characters in option card names.

An alphabetical listing of the available cards follows. The new cards are highlighted with bold type:

ATTLCOE - Set values of attenuation function coefficients.

ATTLOC - Compute location by the new method using attenuation function.

ATTMAG - Compute magnitude by the alternative method using attenuation function.

AZPAR - Set parameters of the azimuth algorithm.

AZVAR (new card) – Assign the azimuth calculated for method 0 to other methods.

BTSTRP - Compute bootstrap uncertainties.

COMMADIVI (new card) - For comma/tab/other separated input file without fixed format.

COMMADIVS (new card) - For comma/tab/other separated input setting file without fixed format.

COMPCATT (new card) - Compute coefficient for magnitude using attenuation function.

COMPCOEF - Compute new coefficients for magnitude-intensity relations (no attenuation).

CRAMCOEF - Set coefficients of the distance weighting formula of the azimuth algorithm.

ELLIPCL - Set confidence level for uncertainty ellipse and/or ellipsoid.

FILE - Set the name of the macroseismic input data file.

FORMAT - Set the Fortran format of the event identification records in the input data file.

FORMATI - Set the Fortran format of the intensity observation records in the input data file.

GMT - Produces files for plotting with GMT.

INPFOLDER - Set the name of the subfolder where input files reside.

INTRANGE (new card) - Defines the usable intensity range.

LENGCOEF - Set the coefficients of the formula to compute the fault length from magnitude.

MAGCOEF - Set the coefficients of the formula to compute magnitude from isoseismal areas.

MAPINFO - Produce the import files for MapInfo.

MUSSONDEP - Compute the source depth using *Musson* (1996) method.

M-I0COEF - Set the coefficients of the magnitude-epicentral intensity formula (old method).

M-IECATT - Set the coefficients of the magnitude-intensity at epicentre formula (new method).

NMNIEPI (new card) - Set the minimum number of intensities to compute the method 0 epicentre.

NOI0 (new card) - Skip I0 for computation of magnitude coefficients (in COMPCATT)

NOHEADER (new card) - Compute parameters of earthquakes without use a header.

OUTFOLDER - Set name of the subfolder where the output files are stored.

OUTLOOK - Create a fixed format summary file for certain applications.

OUTPUT - Set the detail of output in the outsummary.datfile.

PATHCD - Set the root path of input and output files.

POLYW (new card) - Check if computed macroseismic epicentres are offshore or in land.

ROUNDINT (new card) - Defines the rounding of the intensity, leaving any intensities unchanged.

RVLATLON (new card) - Reverse lat/lon order for FORMATI card

SBDEP (new card) - Compute Depth with approach of Sbarra et al., 2019 (**SBDEP19**) or 2023 (**SBDEP23**).

SELAREA (new card) - Identify ore remove earthquakes inside selected areas (max 10 areas).

SELMAG - Set the minimum magnitude for computing fault orientation.

SETGMT - Define the command line to set GMT environment.

SETPOLY (new card) - Set options for reading coastal polylines.

THRINT (new card) - Minimum number of intensities for earthquake to compute parameters.

USEMAG - Use the magnitude reported in the event identification record.

WIDCOEF - Set the coefficients of the formula to compute the fault width from magnitude.

A detailed explanation of the format of each option card is given below.

ATTLCOE

This card sets the values of the coefficients and of the reference depth used the location and magnitude methods based on a log-linear attenuation function (e.g. *Pasolini et al.*, 2008). The first parameter indicates if a cut of the data at long distances is made (S) or not (E) (see *Gasperini et al.*, 2010 for details). If the coefficients and the depth are all zero or blank, those computed by *Pasolini et al.* (2008) or by *Gasperini et al.* (2010) for Italy are used, with or without the long distance cut, depending on the value of first parameter (S or E respectively). In this case, the coefficients of the linear regression between instrumental magnitude and intensity at the epicentre (option card M-IECATT) are also set to the corresponding values reported by *Gasperini et al.* (2010). If the option card is missing, no cut is made and the coefficients computed by *Gasperini et al.* (2010) are used.

Parameter field:

- 1) Name: *Code*
Function: if S a cut of data at long distances is made, if E no cut is made
Columns: 11
Type: Character
Default: E
- 2) Name: *Aattcoe*
Function: Coefficient of the linear term
Columns: 16-25
Type: Real
Default: 0.00091
- 3) Name: *Battcoe*
Function: Coefficient of the logarithmic term
Columns: 26-35
Type: Real
Default: 1.17219
- 4) Name: *Depref*
Function: Reference hypocentral depth
Columns: 36-45
Type: Real
Default: 4.487

Example (or delimiter by COMMADIVIS)

```
1-----11---16-----26-----36-----
ATTLCOE    E          0.00091    1.17219    4.487
```

ATTLOC

When this option card is present, the location is computed (also) by methods based on a log-linear attenuation equation (see *Gasperini et al.*, 2010). The minimum number of intensity data needed to compute the location, and the method (or methods) to employ can be specified optionally. By default, method 4 only is selected. This option is disabled anyhow if COMCOEFF is included in the card deck.

Parameter field:

- 1) Name: *Nmloc*
Function: Minimum number of intensity data needed to compute the location
Columns: 11-15
Type: Integer

Default: 10

2) Name: *ininc*(1)

Function: If not 0 or blank, select location method 1

Columns: 16-20

Type: Integer

Default: 0

3) Name: *ininc*(2)

Function: If not 0 or blank, select location method 2

Columns: 21-25

Type: Integer

Default: 0

4) Name: *ininc*(3)

Function: If not 0 or blank, select location method 3

Columns: 26-30

Type: Integer

Default: 0

5) Name: *ininc*(4)

Function: If not 0 or blank, select location method 4

Columns: 31-35

Type: Integer

Default: 1

6) Name: *ininc*(5)

Function: If not 0 or blank, select location method 5

Columns: 36-40

Type: Integer

Default: 0

7) Name: *ininc*(6)

Function: If not 0 or blank, select location method 6

Columns: 41-45

Type: Integer

Default: 0

Example (or delimiter by COMMADIVIS)

```
1-----11---16---21---26---31---36---41---
ATTLOC      10      0      0      0      1      0      0
```

ATTMAG

When this option card is present, the magnitude is computed (also) by the method based on a log-linear attenuation equation (see *Gasparini et al.*, 2010).

Parameter field:

1) Name: *Nmmag*

Function: Minimum number of intensity data needed to compute the magnitude

Columns: 11-15

Type: Integer

Default: 10

Example (or delimiter by COMMADIVIS)

```
1-----11---
ATTMAG      20
```

AZPAR

This card sets the values of the minimum number of data-points needed to compute the orientation, and of the maximum of half degree decrements with respect to maximum intensity to be applied (see: *Gasperini et al.*, 1999).

Parameter field:

1) Name: *Nmin*

Function: minimum number of data-points needed to compute fault orientation

Columns: 11-15

Type: Integer

Default: 3

2) Name: *Ndecr*

Function: maximum number of half degree decrements from maximum intensity

Columns: 16-20

Type: Integer

Default: 4

Example (or delimiter by COMMADIVIS)

```
1-----11---16---
AZPAR      6      4
```

AZVAR (*new card*)

This card assigns the azimuth calculated for method 0 to other methods. The azimuth calculation is performed using the highest intensities. As, when using methods 1-6, the epicentre might be far from the epicentre of method 0, a clear discrepancy between the datasets used to calculate the epicentre and the azimuth might occur. If the distance between the epicentres of method 0 and other methods 1-6 is larger than the assigned threshold value, the azimuth for the other methods is equated to that of method 0. Moreover, if the epicentre of the alternative method is located offshore, with respect to the polyline (POLYW), the azimuth of method 0 is automatically assigned.

Parameter field:

1) Name: *dist*

Function: distance beyond which to assigns the azimuth of method 0

Columns: 11-20

Type: Real

Default: -

Example (or delimiter by COMMADIVIS)

```
1-----11-----21
AZVAR      1.0
```

BTSTRP

If this card is present, the program computes uncertainties by the bootstrap method. The number of bootstrap replicate sets N_{repl} is computed as a function of the parameters described below and of the number of intensity data points N_{po} of each earthquake as

$$N_{repl} = N_{btstrp} + N_{po} * F_{btstrp}$$

Default values for N_{btstrp} and F_{btstrp} are assumed only if both of them are zero or blank. If only N_{btstrp} is zero or blank, the number of replicate sets is simply proportional to the number of intensity data according to the coefficient F_{btstrp} . If only F_{btstrp} is zero or blank, the number of replicate sets is N_{btstrp} for all the earthquakes.

Parameter field:

1) Name: N_{minpo}

Function: Minimum number of intensity data to compute uncertainties by the bootstrap method

Columns: 11-15

Type: Integer

Default: 20

2) Name: N_{btstrp}

Function: Minimum number of bootstrap replicate sets

Columns: 16-20

Type: Integer

Default: 20

3) Name: F_{btstrp}

Function: Factor multiplying the number of data in the formula of the number of bootstrap sets

Columns: 21-25

Type: Real

Default: 0.5

Example (or delimiter by COMMADIVIS)

1-----11---16---21---

BTSTRP 30 100 0.3

COMMADIVI (new card)

The data input file contains fields separated by a specific delimiter (comma, tab, etc.) instead of fixed width fields according to the FORMATI and FORMATE cards. Note that “T” indicates tab delimited. We advise against the use of a blank character.

The use of COMMADIVI makes virtually unnecessary the FORMATE and FORMATI cards, but the ordering of fields is the same as FORMATE and FORMATI.

Parameter field:

1) Name: *Field delimiter*

Function: the chosen character allows to divide data of input file

Columns: 11-15

Type: Character

Default: Tabulation. No allowed blank character to divide

Example

```
1-----11---15
COMMADIVI      T
```

COMMADIVS (new card)

As for COMMADIVI card for the inpparm.dat file cards.

Parameter field:

1) Name: *Field delimiter*

Function: the chosen character allows to divide the card settings of inpparm.dat file

Columns: 11-15

Type: Character

Default: Tabulation. No allowed blank character to divide

Example

```
1-----11---15
COMMADIVS      T
```

COMPCATT (new card)

This card must be the first one of the card deck. The code run the procedure for re-computing coefficients for magnitude using the attenuation function. If the instrumental magnitude is absent or zero, the event is not considered for regressions. All option cards (including comments) after this card in the inpparm.dat file are copied to the outparm.dat file. The computed regression coefficients are added to the end of the deck in outparm.dat file. Hence, to use the new parameters in further computations, the file outparm.dat can be simply renamed as inpparm.dat. See also SELAREA card to exclude certain areas (e.g. volcanic areas). Note that COMPCATT disables ATTLOC, ATTMAG and BSTRP.

Parameter field:

1) Name: *Nmm*

Function: minimum number of intensity observations needed for each intensity class used in regressions

Columns: 11-15

Type: Integer

Default: 5

Example

```
1-----11---
COMPCOA        3
```

COMPCOEF

This card must be inserted at the beginning of the card deck. The program must run the procedure to computing the coefficients of magnitude-intensity regressions, on the basis of instrumental magnitudes (taken from the event identification records) associated with macroseismic observations. If the

magnitude is absent or zero, the event is not considered for regressions. All option cards after this card in the inpparm.dat file, excepting MAGCOEF and M-I0COEF but including comments, are copied to the outparm.dat file. At the end of computations, new MAGCOEF and M-I0COEF cards reporting the computed regression coefficients are added to the end of the deck in the outparm.dat file. Hence, to use the new parameters in further computations, the file outparm.dat can be simply renamed as inpparm.dat. See also SELAREA card to exclude certain areas (e.g. volcanic areas). Note that COMPCOEF disables ATTLOC, ATTMAG and BSTRP. In this release COMPCOEF was modified to potentially eliminate I0 from the calculation of magnitude coefficients (see NOI0 card and *Vannucci et al.*, 2024).

Parameter field:

1) Name: *Nmm*

Function: minimum number of intensity observations needed for each intensity class used in regressions

Columns: 11-15

Type: Integer

Default: 5

Example

```
1-----11---
COMPCOEF      3
```

CRAMCOEF

This card sets the values to the coefficients of the *CRAM* (*Berardi et al.*, 1993) formula to compute the distance weights used by the algorithm to determine fault orientation (see *Gasperini et al.*, 1999; 2010)

$$\Delta I = a + b^3 \sqrt{D}$$

where ΔI is the difference between epicentral and local intensity and D is the epicentral distance. Default values were empirically estimated for Italy by *Gasperini et al.* (1999).

Parameter field:

1) Name: *Acram*

Function: Intercept of the CRAM equation

Columns: 11-20

Type: Real

Default: -0.46

2) Name: *Bcram*

Function: Coefficient of the *CRAM* equation

Columns: 21-30

Type: Real

Default: 0.93

Example

```
1-----11-----21---
CRAMCOEF  -0.3      1.1
```

ELLIPCL

This card sets the confidence level ($0 < c.l. < 1$) at which the semiaxes of uncertainty ellipses and/or ellipsoids are to be computed. If the parameter is ≥ 1 (and < 5) it is assumed as the factor by which the standard (1σ) semiaxes are to be multiplied. In this case the confidence levels are different for ellipses and ellipsoids. For example, if the parameter is 1 the confidence level is $c.l.=0.3935$ for ellipses and $c.l.=0.1987$ for ellipsoids while if the parameter is 2, $c.l.=0.8647$ for ellipses and $c.l.=0.738$ for ellipsoids.

Parameter field:

1) Name: *Elcl*

Function: Confidence level or multiplying factor for uncertainty ellipses and ellipsoids

Columns: 11-20

Type: Integer

Default: 1.0

Example

```
1-----11-----
ELLIPCL          0.90
```

FILE

This card sets the name of the intensity data file. The file must reside in the same folder of the program executable or in the folder specified by the INPFOLDER card. For each earthquake, an identification header record must precede the intensity observation records unless the NOHEADER card is included in the card deck. The Fortran format can be specified by FORMATE and FORMATI option cards.

Parameter field:

1) Name: *Filename*

Function: name of the input data file

Columns: 11-80

Type: Character

Default: input.dat

Example (or delimiter by COMMADIVIS)

```
1-----11-----21----- ... 71-----
FILE          input2.dat
```

FORMATE

This card sets the Fortran format (embedded in parenthesis) of the event identification records in the input data file. It must contain 9 or 11 fields (depending on the type of computations) with the following ordering: Year (integer), Month (integer), Day (integer), Hour (integer), Minute (integer), Second (integer), Epicentral Area (character), Magnitude (real), Number of intensity points (integer), Standard error of Magnitude (real), Epicentral intensity (real). The last two fields are required only for the computation of magnitude-intensity coefficients (see COMPCOEF card). "T" fields can be used if the ordering of fields in the user record is different (see example).

Parameter Field:

1) Name: *Formate*

Function: Fortran format of event summary records

Columns: 1-80

Type: Alphanumeric

Default: -

Example (or delimiter by COMMADIVIS):

```
1-----11-----21-----31----- ... 71-----
FORMATE      (8x,I5,5I3,1X,A20,64x,F3.2,T58,I4,1x,f4.0,1x,f4.0)
```

FORMATI

This card sets the Fortran format (embedded in parenthesis) of the intensity observation records in the input data file. It must contain 4 fields in the following order: Latitude (real), Longitude (real), Intensity (real), Locality name (alphanumeric). Tabulation (T fields) can be used if the ordering of fields in the user records is different. Latitude and longitude must be given in decimal degrees (not primes). Westward longitudes and southward latitudes can be specified with a negative sign. Uncertain intensity values (i.e. VII-VIII) may be given as semi integer values (i.e. 7.5). Other unconventional intensity values (e.g. 9.1, 3.9, 4.6, 8.6) are accepted but their specific treatment is at user risk (see also ROUNDINT card). Latitude and longitude can be changed with RVLATLON card. "T" fields can be used if the ordering of fields in the user record is different (see example).

Parameter Field:

1) Name: *Formati*

Function: Fortran format of intensity data records

Columns: 1-80

Type: Alphanumeric

Default: (2F7.3,1X,F4.1,1X,A20)

Example (or delimiter by COMMADIVIS):

```
1-----11-----21-----31----- ... 71-----
FORMATI      (48X,2F8.4,F6.1,T30,A20)
```

GMT

When this option card is present, the program provides the files for plotting epicentres, boxes etc. using GMT (*Wessel and Smith*, 1991; 1998). The first 2 binary digits of the parameter activate (1) or deactivate (0) the production of summary files with epicentres (epi_mmethod_input.gmt and .bat) as well as summary files with boxes (box_mmethod_input.gmt and .bat), if the orientation is computed or circles (cir_mmethod_input.gmt and .bat) if not. They are stored in the subfolder GMT\SUMMARY of the output folder. The last 3 digits activate the production of individual files for each earthquake with epicentres (ep_mmethod_year_month_day_area.gmt and .bat), boxes or circles (bx_mmethod_year_month_day_area.gmt and .bat or ci_mmethod_year_month_day_area.gmt and .bat) or intensity observations (qt_mmethod_year_month_day_area.gmt and .bat) that are stored in subfolders GMT\INDIVIDUAL\year_month_day_area subfolders of the output folder. In file or folder names, *method* is the location method number (0-6), *input* is the file name specified in the FILE option

card, *year*, *month*, *day* and *area* are the year, month, day and epicentral area respectively reported in the event identification record for the given earthquake.

Parameter field:

- 1) Name: *Iexcl*
 Function: binary digits activating various GMT files
 Columns: 11-15
 Type: Integer
 Default: 00000

Example (or delimiter by COMMADIVIS)

```
1-----11---
GMT      11000
```

INPFOLDER

This card sets the name of the folder where input files reside. The name must be a valid folder name for the operating system used and must not include blank characters. If the card is absent, the input file is assumed to reside in the same folder of the executable file.

Parameter field:

- 1) Name: *Foldername*
 Function: Name of the subfolder containing the input data file
 Columns: 11-80
 Type: Character
 Default: FILEINPUT

Example (or delimiter by COMMADIVIS)

```
1-----11-----21----- ... 71-----
FILE      input_data_folder
```

INTRANGE (*new card*)

This card allows to limit the range of intensities to be processed, discarding values below or above the limits.

Parameter Field:

- 1) Name: *lower boundary*
 Function: lower intensity value to be processed
 Columns: 11-20
 Type: Real
 Default: 2.0
- 2) Name: *higher boundary*
 Function: higher intensity value to be processed
 Columns: 11-20
 Type: Real
 Default: 12.0

Example (or delimiter by COMMADIVIS)

```
1-----11-----21-----
INTRANGE          2.0      12.0
```

LENGCOEF

This card sets the values of the coefficients of the formula used to compute fault length L (in km) from moment magnitude M_w

$$\text{Log}_{10}(L) = a + bM_w$$

Default values are the empirical coefficients of the subsurface rupture length (RLD) for “all types of faults” computed by *Wells and Coppersmith* (1994).

Parameter field:

- 1) Name: *Alen*
 Function: Intercept of the fault length formula
 Columns: 11-20
 Type: Real
 Default: -2.44
- 2) Name: *Blen*
 Function: Coefficient of the fault length formula
 Columns: 21-30
 Type: Real
 Default: 0.59

Example (or delimiter by COMMADIVIS)

```
1-----11-----21-----
LENGCOEF          -2.11      0.65
```

MAGCOEF

This card sets the values of the coefficients, for different intensity class, of the *Sibol et al* (1987) formula to compute the macroseismic magnitude as a function of the isoseismal areas and of epicentral intensity

$$M_i = a + b\text{Log}^2(A_i) + c(I_0)^2$$

where A_i is the isoseismal area for the i -th intensity class, which is computed as

$$A_i = \pi(R_i)^2$$

where R_i (in Km) is the average epicentral distance of localities belonging to the i -th intensity class. The magnitude of an earthquake is computed as the weighted average of the single estimates made using the intensity classes available for the given earthquake. If the number of intensity classes is 4 or larger the largest and lowest estimates are trimmed off the averaging. Two different methods can be selected to weight the estimates made by different intensity classes. For the older one (the only one available in the oldest versions of Boxer) the weight is simply assumed as inversely proportional to the square of the logarithm of the number of available intensity observations. For the newer method, the weight is inversely proportional to the number of intensity observations and to the square of

regression standard deviation for the corresponding intensity class (computed during the coefficient computation run selected by COMPCOEF card). For the older method, only the values of the 3 coefficients of the *Sibol et al.* (1987) formula and the limits of intensity classes are needed on data cards described below. For the new method, three additional parameters are needed for each intensity class, the standard deviation of the regression, the weight normalization factor (corresponding to the number of intensity observation for a unit weight) and the number of degrees of freedom of the regression (the number of data minus the number of free parameters). In absence of a MAGCOEF option card or if the third parameter is zero or blank for at least an intensity class, the older weighting method is used. Default values are empirical estimates made for Italy from the data of the *Catalogo dei Forti Terremoti in Italia dal 461 a.C. al 1990* (2) (*Boschi et al.*, 1997) in terms of moment magnitude.

Values reported below refer to the calibration proposed in *Vannucci et al.*, 2026

Parameter field:

1) Name: *Ncoef*

Function: number of intensity classes used to compute magnitude

Columns: 11-15

Type: Integer

Default: 10

Additional data cards Card 1

1) Name: *Aiv(1)*

Function: Lower limit of the intensity class (the higher limit is specified in the next card)

Columns: 1-15

Type: real

Default: see below

2) Name: *Coef(1,1)*

Function: *a* coefficient of the *Sibol et al* (1987) formula

Type: real

Columns: 16-30

Default: see below

3) Name: *Coef(1,2)*

Function: *b* coefficient of the *Sibol et al* (1987) formula

Type: real

Columns: 31-45

Default: see below

4) Name: *Coef(1,3)*

Function: *c* coefficient of the *Sibol et al* (1987) formula

Type: real

Columns: 46-60

Default: see below

5) Name: *Stdr(1)*

Function: standard deviation of the regression (for new weighting method only)

Type: real

Columns: 61-68

Default: see below

6) Name: *Avme(1)*

Function: weight normalization factor (for new weighting method only)

Type: real

Columns: 69-76

Default: see below

7) Name: *Idfe(1)*

Function: degrees of freedom of regression (for new weighting method only)

Type: integer

Columns: 77-80

Default: see below

Cards 2 to *Ncoef*

Same as Card 1 for different intensity classes (with increasing intensity).

Example (or delimiter by COMMADIVIS):

```

1-----15-----//-----80
MAGCOEF      1 6
      2.0      3.14625      0.05502      0.01697  0.2996 12.0084 116
      2.5      3.09158      0.05419      0.01794  0.3020 12.9634  79
      3.0      3.12116      0.05609      0.01892  0.2969 23.5970 265
      3.5      3.17512      0.05301      0.02110  0.2916 21.9126 180
      3.9      3.13565      0.05301      0.02235  0.2844 13.2000  47
      4.0      3.14411      0.06224      0.01987  0.2412 27.2061 293
      4.5      3.10739      0.06639      0.02191  0.2330 25.1632 187
      5.0      3.38589      0.06138      0.02040  0.2460 27.4221 196
      5.5      3.77330      0.04479      0.01995  0.2602 21.6667  81
      6.0      3.79646      0.05817      0.01843  0.2399 29.2927  79
      6.5      4.05438      0.04829      0.01775  0.2336 25.5686  48
      7.0      3.94310      0.06161      0.01822  0.2116 38.8571  39
      7.5      4.31152      0.03646      0.01800  0.1427 26.9091  19
      8.0      4.54605      0.05705      0.01451  0.1140 40.0556  15
      8.5      4.71048      0.04997      0.01460  0.0673 29.1429   4
      9.0      5.61608      0.14427      0.00000  0.1445 23.1429   5

```

MAPINFO

When this option card is present, the program produces the import files (.mif and .mid) for displaying epicentres, boxes etc., using MapInfo. The first 2 binary digits of the parameter activate (1) or deactivate (0) the production of summary files with epicentres (epi_mmethod_input.mif and .mid) as well as summary files with boxes (box_mmethod_input.mif and .mid) if the orientation is computed, or circles (cir_mmethod_input.mif and .mid) if not. They are stored in subfolder MAPINFO\SUMMARY of the output folder. The last 3 digits activate the production of individual files for each earthquake with epicentres (ep_mmethod_year_month_day_area.mif and .mid), boxes or circles (bx_mmethod_year_month_day_area.mif and .mid or ci_mmethod_year_month_day_area.mif and .mid) or intensity observations (qt_mmethod_year_month_day_area.mif and .mid) that are stored in subfolders MAPINFO\INDIVIDUAL\year_month_day_area subfolders of the output folder. In file or folder names, method is the location method number (0-6), input is the file name specified in the FILE option card, year, month, day and area are the year, month, day and epicentral area respectively reported in the event identification record for the given earthquake.

Parameter field:

1) Name: *Iexcl*

Function: binary digits activating various MapInfo import files

Columns: 11-15

Type: Integer

Default: 00000

Example (or delimiter by COMMADIVIS)

1-----11---

MAPINFO 11000

MUSSONDEP

If this option card is present, the program computes the depth by the *Musson* (1996) method using the parameters indicated in the parameter field or using default values, if zero or blank.

Parameter field:

1) Name: *Alpha*

Function: Attenuation constant

Columns: 11-20

Type: Real

Default: 0.0061

1) Name: *Gamma*

Function: Coefficient relating intensity to physical ground motion

Columns: 21-30

Type: Real

Default: 3.2761

Example (or delimiter by COMMADIVIS)

1-----11-----21-----

MUSSONDEP 0.003 3.0

M-I0COEF

This card sets the values of the coefficients of the formula to compute macroseismic magnitude M from epicentral intensity I_0 .

$$M = a + bI_0$$

When the intensity data are insufficient to apply the isoseismal areas method (see: *Gasperini et al.* (1999)) this value is reported in the outsummary.dat file. Default values are calibrated for moment magnitude (*Hanks and Kanamori*, 1979) and have been empirically derived by combining the M_s - I_0 relation estimated by *Rebez and Stucchi* (1999) for the Italian magnitude-intensity database

$$M_s = 0.94(\pm 0.13) + 0.56(\pm 0.017)I_0$$

with the Log M_0 - M_s relation (in dyne cm) estimated for Italy and surrounding areas by *Gasperini and Ferrari* (2000)

$$\text{Log } M_0 = 19.3(\pm 0.3) + 0.96(\pm 0.06)M_s = 20.20 + 0.538I_0$$

and then with the definition of M_w by *Hanks and Kanamori* (1979)

$$M_w = -10.7 + 2/3 \text{Log } M_0 = 2.768 + 0.3584 I_0$$

Parameter field:

1) Name: *Aint*

Function: intercept of the M-I0 equation

Columns: 11-20

Type: Real

Default: 2.768

2) Name: *Bint*

Function: coefficient of the M-I0 equation

Columns: 21-30

Type: Real

Default: 0.3584

Example (or delimiter by COMMADIVIS)

```
1-----11-----21-----
M-I0COEF      1.5      0.5
```

M-IECATT

This card sets the values of the coefficients of the formula to compute macroseismic magnitude M from intensity at the epicentre I_E by the method proposed by *Gasparini et al.* (2010)

$$M = a + b I_E$$

They are also set by the option card ATTLCOE, according to the value of *Code* (S or E), when the coefficients of attenuation equation are set to default values for Italy.

Parameter field:

1) Name: *Amg*

Function: intercept of the M-IE equation

Columns: 11-20

Type: Real

Default: 5.368/2.3638

2) Name: *Bmg*

Function: coefficient of the M-Ie equation

Columns: 21-30

Type: Real

Default: 1./2.3638

Example (or delimiter by COMMADIVIS)

```
1-----11-----21-----
M-IECATT      2.267      0.423
```

NMNIEPI (*new card*)

This card allows to set the minimum number of intensities to compute the epicentre for the barycentre

method (or method 0). Note that for alternative methods (1-6) the minimum number of intensities is defined in ATTLOC.

Parameter field:

1) Name: *ival*

Function: minimum number of intensities

Columns: 11-15

Type: Integer

Default: 3

Example (or delimiter by COMMADIVIS)

```
1-----11---
NMNIEPI      3
```

NOI0 (*new card*) -

This card allows I0 to be eliminated from the calculation of magnitude coefficients in COMPCOEF (see card). The card can be used for catalogues where I0 is not present (see *Vannucci et al.*, 2024).

Example (only card, also for COMMADIVIS)

```
1-----11
NOI0
```

NOHEADER (*new card*)

This card allows to process earthquake intensity data without a header with information such as event date or intensity number. It can only be used for one earthquake at a time, but it can be useful for quickly calculating the parameters of MDP distribution. For example, it could be applied when intensities are available in near real time (*Vannucci et al.*, 2024) and for one earthquake only.

Example (only card, also for COMMADIVIS)

```
1-----11
NOHEADER
```

OUTFOLDER

This option card sets the name of the folder where output files are to be stored. The name must be a valid folder name for the operating system used and must not include blank characters. If the card is absent, the output files are stored in the “current directory”, i.e. at the path from which the executable programme is run.

Parameter field:

1) Name: *Foldername*

Function: Name of the subfolder containing the input data file

Columns: 11-80

Type: Character

Default: -

Example (or delimiter by COMMADIVIS)

```
1-----11-----21----- ... 71-----
FILE      output_data_folder
```

OUTLOOK (undocumented)

When this option card is present, the program produces a fixed-format summary file used by some applications.

OUTPUT

This card set the detail of output (=1 normal output, >1 long output) in the outfull.dat file. Parameter field:

- 1) Name: *Iout*
 Function: type of output
 Columns: 11-15
 Type: Integer
 Default: 1

Example (or delimiter by COMMADIVIS)

```
1-----11----
OUTPUT      2
```

PATHCD

This option card sets the root device or folder containing input and output files Parameter field:

- 1) Name: *Foldername*
 Function: Name of the device or subfolder
 Columns: 11-80
 Type: Character
 Default: None (current folder)

Example (or delimiter by COMMADIVIS)

```
1-----11-----21----- ... 71-----
PATHCD     C:\boxerfolder
```

POLYW (*new card*)

This card allows to use coastal polylines to determine if computed epicentres are offshore or inland. Temporary files are also used for the calculation. The polylines contain 1) a header with a symbol indicating the start of a new polyline, 2) longitude and latitude coordinates (in the order indicated) separated by empty spaces or tab or comma (see SETPOLY card). For each earthquake, it is assessed whether the macroseismic epicentre is within the polylines, i.e. inland, or not (offshore), providing a

flag (F or T, respectively) in the output files. The maximum number of areas to check is 33,000. Note that POLYW is disabled for bootstrap datasets: no calculations are performed for resampling datasets. To use this card, SETPOLY has to be set.

Parameter field:

1) Name: *name of polyline file*

Function: Open coastal polyline to check if earthquake is offshore or inland.

Columns: 11-15

Type: Character

Default: *empty field. User must specify the file.*

Example (or delimiter by COMMADIVIS)

```
1-----11-----21----- ... 71-----
```

```
POLYW      C:\coastal_line\Italy.dat
```

ROUNDINT (*new card*)

This card allows to round the intensities to the nearest half (or integer) degree for unconventional intensities (values such as 6.4, 8.6, etc, see *Vannucci, 2025*). Two values of intensity (4.6 and 3.9) only are always excluded from the rounding procedure which were used in the Italian catalogues to indicate a generic “felt”.

This card allows to round the intensity: *i*) to the nearest half (or integer) degree (*rounding type*: “R”, e.g. 8.6 becomes 8.5), *ii*) to the higher half (or integer) degree (*rounding type*: “+”, e.g. 8.6 becomes 9.0), *iii*) to the higher half (or integer) degree (*rounding type*: “-”, e.g. 6.4 becomes 6.0) and *iv*) to discard unconventional intensities (*rounding type*: “E”, e.g. 6.4 becomes unusable).

Parameter field:

1) Name: *rounding type*

Function: Round intensity to nearest integer degree or half degree value (R), higher (+) or lower (-) integer degree. To discard anomalous intensities types “E”

Columns: 11

Type: Character

Default: R. Examples: “R”, “+”, “-“, “E”

2) Name: *n. of excluded intensity*

Function: N. of other intensities that have not to be rounded

Columns: 12-15

Type: Integer

Default: -

3) Name: *values of intensity excluded*

Function: intensity values excluded from the rounding procedure

Columns: 16- every 5 characters

Type: Real

Default: -

Example (or delimiter by COMMADIVIS)

```
1-----11-----21----- ... 71-----
```

```
ROUNDINT  R      2    6.4    8.6
```

RVLATLON

This card allows to invert the ordering of coordinates in FORMATI. FORMATI reads in sequential longitude and latitude order. If the original file provides latitude and longitude, this cards reverses the coordinates.

Example (only card, also for COMMADIVIS)

```
1-----11
```

```
RVLATLON
```

SBDEP (new card) i.e. SBDEP19 or SBDEP23

This card allows to apply the method of *Sbarra et al.*, 2019 or 2023 (SBDEP19 or SBDEP23 card, respectively) to compute the macroseismic depth and magnitude. Starting from the computed macroseismic epicentre for all the BOXER methods, the code: 1) computes the distance of each MDP with respect to epicentre; 2) selects the MDPs within a distance range; 3) sorts the localities (i.e. intensities) in function of distance from epicentre; 4) computes the slope distance-intensity by using selected MDPs.

Although parameters such as the distance within which to select MDPs are defined in the approach of *Sbarra et al.* (2019, 2023), the card offers the options to vary some parameters in distance and set the minimum number of MDPs inner the range of distance to compute the slope.

Notes: No options by the User is equivalent to using the default parameters.

Parameter field:

1) Name: *Dstart*

Function: min distance with respect to macroseismic epicentre to compute the slope of MDPs

Columns: 11-20

Type: Real

Default: 0

2) Name: *Dend*

Function: max distance with respect to macroseismic epicentre to compute the slope of MDPs

Columns: 21-30

Type: Real

Default: 55

3) Name: *Ndlim*

Function: minimum number of MDPs to compute regression Distance-Intensity

Columns: 31-40

Type: Real

Default: 20

Example (or delimiter by COMMADIVIS)

```
1-----11-----21-----31-----41
```

```
SBDEP23      0          55          60
```

SELAREA (new card)

This card allows to add a Flag (E) in the output files, setting “selected” areas (until max number of 10 areas), by reference coordinates and radius. Earthquakes inner the radius of each area are indicated as “E”. By using with COMCATT/COMPCOEFF the earthquakes located inner these areas are excluded from computation. For example, the user could define volcanic areas and not use earthquakes within

them. Note that this function cannot solve more complex geometric problems involving data preparation and selection.

Parameter field:

1) Name: *header (polyline of area)*

Function: n. of areas. This defines a loop to read the info of each area (latitude, longitude, radius)

Columns: 11-15

Type: Integer

Default: no default

Then: read for n (number of areas) latitude, longitude, radius of each area, e.g.:

2) Name: area latitude

Function: latitude of the first area

Columns: 16-24

Type: Real

Default: no default

3) Name: area longitude

Function: longitude of the first area

Columns: 25-33

Type: Real

Default: no default

4) Name: area radius

Function: radius of the first area

Columns: 34-42

Type: Real

Default: no default

....Then: repeat latitude, longitude, radius for other areas (max 10 areas)...

Example (or delimiter by COMMADIVIS)

```
1-----11-15----- 24-----33-----42 ... repeat for n areas
SELAREA          1      37.00      15.00      25.00
```

SELMAG

When this option card is present, the program computes the azimuth only for earthquakes with magnitude not lower than the given threshold.

Parameter field:

4) Name: *Aminmg*

Function: minimum magnitude threshold above which the azimuth is computed

Columns: 11-15

Type: Real

Default: none (no threshold)

Example (or delimiter by COMMADIVIS)

```
1-----11-----
SELMAG          5.5
```

SETGMT

This option card defines the shell command to set the environment variables for GMT (complete path to the `gmtenv` file). This line is reported at the beginning of each `.sh` or `.bat` file produced by the program to plotting results by GMT (see option card GMT).

Parameter field:

- 1) Name: *phgm*
 Function: shell command line
 Columns: 11-80
 Type: Character
 Default: `gmtenv.bat`

Example (or delimiter by COMMADIVIS)

```
1-----11-----21-----31----- ... 71-----
SETGMT      C:\gmt\src\gmtenv.bat
```

SETPOLY (*new card*)

This card defines the options for reading coastal polylines, i.e. land areas within a coastline (see POLYW card). If no *delimiter* is defined (see below), longitude and latitude must be separated by empty space. Note that a tab separator is indicated as “T” in the *delimiter*.

Parameter field:

- 1) Name: *header (polyline of area)*
 Function: symbol marking the start of a new polyline delimiting an area of land
 Columns: 11-20
 Type: Character
 Default: ‘>’
- 2) Name: *delimiter*
 Function: symbol (e.g. tab, comma) marking the start of a new polyline delimiting an area of land
 Columns: 20-30
 Type: Character
 Default: ‘ ’
- 3) Name: *coordinates order*
 Function: to reverse the order of coordinates (i.e. read latitude and longitude) symbol (:)
 Columns: 30-40
 Type: Character
 Default: ‘ ’, i.e. (default reading: longitude and latitude)

Example (or delimiter by COMMADIVIS)

```
1-----11-----21-----31----- ... 71-----
SETPOLY      >          T
```

THRNINT (*new card*)

This card allows to set the minimum number of intensities to compute parameters of earthquakes. The parameters are calculated only for earthquakes with a number of useful MDPs (e.g. in the selected intensity range) greater than or equal to the set threshold value.

Parameter field:

1) Name: *number of intensities*

Function: Set the minimum intensity number for calculating seismic parameters.

Columns: 11-15

Type: Character

Default: 5

Example (or delimiter by COMMADIVIS)

1-----11----

THRNINT 5

USEMAG

When this option card is present, the program uses the instrumental magnitude reported in the event identification record to compute the length and the width of the fault, in place of the computed value.

Parameter field:

none

Example (or delimiter by COMMADIVIS)

1-----

USEMAG

WIDCOEF

This card sets the values of the coefficients of the formula to compute fault width W from magnitude M

$$\text{Log}_{10}(W) = a + bM$$

Default values are the empirical coefficients of the downdip rupture width (RW) for “all types of faults” computed by *Wells and Coppersmith* (1994).

Parameter field:

1) Name: *Alen*

Function: a coefficient of the fault width formula

Columns: 11-20

Type: Real

Default: -1.01

2) Name: *Blen*

Function: b coefficient of the fault width formula

Columns: 21-30

Type: Real

Default: 0.32

Additional data cardsNone

Example (or delimiter by COMMADIVIS)

1-----11-----21-----

WIDCOEF -1.3 0.29

Format of outsummary_new.dat and outsummary_new.xls files

Such files report the 85 fields listed below as formatted text and tab-delimited text files respectively. The first record contains a header with the synthetic names of the fields. A value of 999 indicates that the parameter is unknown. “Columns” indicates the first and last columns of the field in the outsummary_new.dat.

Name	Type	Columns	Description
ID	Integer	1-6	Earthquake identification number
MT	Integer	9	Location method (0-6)
RLAW	Character	11	Long distance cut code (S or E)
YEAR	Integer	13-16	Year
MONTH	Integer	18-19	Month
DAY	Integer	21-22	Day
HOUR	Integer	25-26	Hour
MINUTE	Integer	28-29	Minute
ES	Character	31	Default: “-“. Offshore (T), inland (F), excluded areas (E)
LAT	Real	35-42	Epicentral latitude (in decimal degrees)
LAT_ERH	Real	44-51	Latitude formal uncertainty (in km)
LAT_BST	Real	53-60	Latitude bootstrap uncertainty (in km)
LON	Real	62-69	Epicentral longitude (in decimal degrees)
LON_ERH	Real	71-78	Longitude formal uncertainty (in km)
LON_BST	Real	80-87	Longitude bootstrap uncertainty (in km)
DEPTH	Real	89-96	Hypocentral depth (in km) ¹
DEP_ERH	Real	98-105	Depth formal uncertainty (in km)
DEP_BST	Real	107-114	Depth bootstrap uncertainty (in km)
TL			Unused
ME	Real	117-122	Preferred magnitude (among old, alternative and I0 methods) ²
ME_ERH	Real	124-129	Preferred magnitude formal uncertainty
ME_BST	Real	131-136	Preferred magnitude bootstrap uncertainty
TM	Character	138	Preferred magnitude type (n, o, a, i) ³
LOCALITY	Character	140-179	Epicentral area geographic indication
NTOT	Integer	181-185	Number of intensity data
NAZ	Integer	187-191	Number of intensity data used to compute fault orientation
NBSTR	Integer	193-198	Number of bootstrap replicate sets
AZIM	Real	201-206	Fault orientation (in degrees)
A_ERR	Real	208-212	Fault orientation formal uncertainty (in degrees)
A_BST	Real	214-286	Fault orientation bootstrap uncertainty (in degrees)
RAYLEIGH	Real	220-227	Rayleigh test significance level
KUIPER	Real	229-236	Kuyper test significance level
I0	Real	238-242	Epicentral intensity I0 (method 0) or intensity at epicentre IE (methods >0)
I_ERH	Real	244-248	Formal uncertainty of I0 or IE
I_BST	Real	250-254	Bootstrap uncertainty of I0 or IE
IM	Real	256-260	Maximum intensity
LENGTH	Real	262-269	Fault length
LEN_BST	Real	271-278	Fault length bootstrap std

WIDTH	Real	280-287	Fault width
WID_BST	Real	289-296	Fault width bootstrap std
CONF_LEVEL	Real	298-307	Confidence level or factor ⁴ of uncertainty ellipses or ellipsoids
EL_AXMX	Real	309-316	Semi-major-axis of formal uncertainty ellipsoid (in km)
EL_AZMX	Real	318-325	Azimuth of major-axis of formal uncertainty ellipsoid (in degrees)
EL_PLMX	Real	327-334	Plunge of major-axis of formal uncertainty ellipsoid (in degrees)
EL_AXMD	Real	336-343	Semi-medium-axis of formal uncertainty ellipsoid (in km)
EL_AZMD	Real	345-352	Azimuth of medium-axis of formal uncertainty ellipsoid (in degrees)
EL_PLMD	Real	354-361	Plunge of medium-axis of formal uncertainty ellipsoid (in degrees)
EL_AXMI	Real	363-370	Semi-minor-axis of formal uncertainty ellipsoid (in km)
EL_AZMI	Real	372-379	Azimuth of minor-axis of formal uncertainty ellipsoid (in degrees)
EL_PLMI	Real	381-388	Plunge of minor-axis of formal uncertainty ellipsoid (in degrees)
EL_BAXMX	Real	390-397	Semi-major-axis of bootstrap uncertainty ellipsoid (in km)
EL_BAZMX	Real	399-406	Azimuth of major-axis of bootstrap uncertainty ellipsoid (in degrees)
EL_BPLMX	Real	408-415	Plunge of major-axis of bootstrap uncertainty ellipsoid (in degrees)
EL_BAXMD	Real	417-424	Semi-medium-axis of bootstrap uncertainty ellipsoid (in km)
EL_BAZMD	Real	426-433	Azimuth of medium-axis of bootstrap uncertainty ellipsoid (in degrees)
EL_BPLMD	Real	435-442	Plunge of medium-axis of bootstrap uncertainty ellipsoid (in degrees)
EL_BAXMI	Real	444-451	Semi-minor-axis of bootstrap uncertainty ellipsoid (in km)
EL_BAZMI	Real	453-460	Azimuth of minor-axis of bootstrap uncertainty ellipsoid (in degrees)
EL_BPLMI	Real	462-469	Plunge of minor-axis of bootstrap uncertainty ellipsoid (in degrees)
E2_AXMX	Real	471-478	Semi-major-axis of formal uncertainty ellipse (in km)
E2_AZMX	Real	480-487	Azimuth of major-axis of formal uncertainty ellipse (in degrees)
E2_AXMI	Real	489-496	Semi-minor-axis of formal uncertainty ellipse (in km)
E2_AZMI	Real	498-505	Azimuth of minor-axis of formal uncertainty ellipse (in degrees)
E2_ECC	Real	507-514	Eccentricity of formal uncertainty ellipse
E2_BAXMX	Real	516-523	Semi-major-axis of bootstrap uncertainty ellipse (in km)
E2_BAZMX	Real	525-532	Azimuth of major-axis of bootstrap uncertainty ellipse (in degrees)
E2_BAXMI	Real	534-541	Semi-minor-axis of bootstrap uncertainty ellipse (in km)
E2_BAZMI	Real	543-550	Azimuth of minor-axis of bootstrap uncertainty ellipse (in degrees)

E2_BECC	Real	552-559	Eccentricity of bootstrap uncertainty ellipse
AATT	Real	561-570	Linear term coefficient of attenuation equation ¹
AATT_ERH	Real	572-581	Formal uncertainty of linear term coefficient
BATT	Real	583-592	Logarithmic term coefficient of attenuation equation ¹
BATT_ERH	Real	594-603	Formal uncertainty of logarithmic term coefficient
MM	Real	605-610	Macroseismic magnitude (old method)
MM_ERHM1	Real	612-617	Old magnitude type i formal uncertainty
MM_ERHM2	Real	619-624	Old magnitude type ii formal uncertainty
MM_BSTM	Real	626-631	Old magnitude bootstrap uncertainty
MT	Real	633-638	Macroseismic magnitude from I0
MT_ERHM	Real	640-645	I0 magnitude formal uncertainty
MT_BSTM	Real	647-652	I0 magnitude bootstrap uncertainty
MM2	Real	654-659	Macroseismic magnitude (alternative method)
M2_ERHM	Real	661-666	Alternative magnitude formal uncertainty
M2_BSTM	Real	668-673	Alternative magnitude bootstrap uncertainty
MS	Real	675-680	Instrumental magnitude
MS_ERHM	Real	682-687	Instrumental magnitude uncertainty

Notes:

¹Reported depth and attenuation coefficients reflect the assumed default values for location method that do not compute them.

²Preferred magnitude is chosen depending on the availability of the estimates made by various methods, using the following order of priority: i) old method by modified *Sibol et al.* (1987) formula, ii) alternative method using the attenuation equation, iii) empirical regression with epicentral intensity.

³*n* stands for the old method with new weighting scheme, *o* for the old method with old weighting scheme, *a* for the alternative method, and *i* for the regression with epicentral intensity.

⁴If the reported value is lower than 1.0, it indicates the confidence level at which uncertainty ellipses and ellipsoids are computed. If instead the reported value is larger or equal 1.0 it indicates the factor by which standard (1σ) ellipse and ellipsoid semiaxes are multiplied.

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